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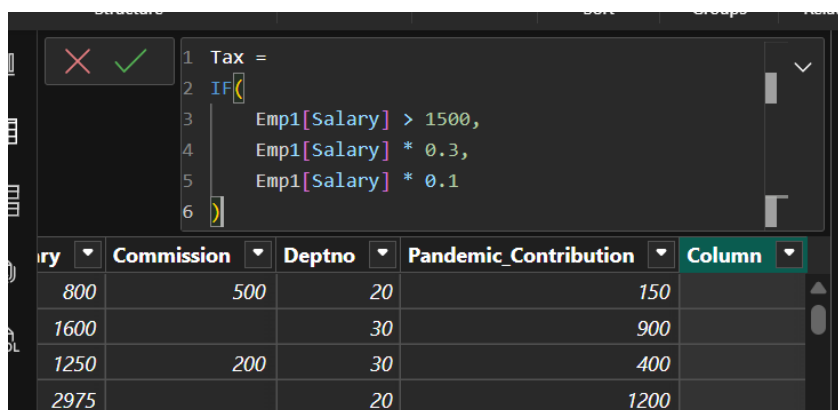
Class : AIML B1

PRN : 23070126095

(The document has both DAX and Measure Questions)

PART A - DAX Queries

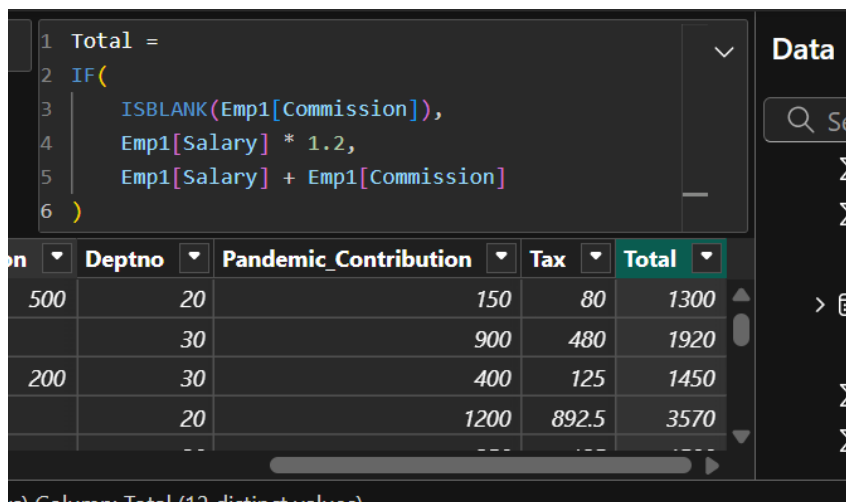
1. Create a column Tax which will show tax value as 30% of Salary for employees earning salary above 1500. For employees earning salary less than or equal to 1500 the tax value shown should be 10% of their salary.



```
1 Tax =
2 IF(
3     Emp1[Salary] > 1500,
4     Emp1[Salary] * 0.3,
5     Emp1[Salary] * 0.1
6 )
```

Salary	Commission	Deptno	Pandemic_Contribution	Tax
800	500	20	150	80
1600		30	900	480
1250	200	30	400	125
2975		20	1200	892.5

2. Create a Total Column which will be Salary plus Commission. If the Commission is not provided then add 20% of Salary in the Salary as the Total.

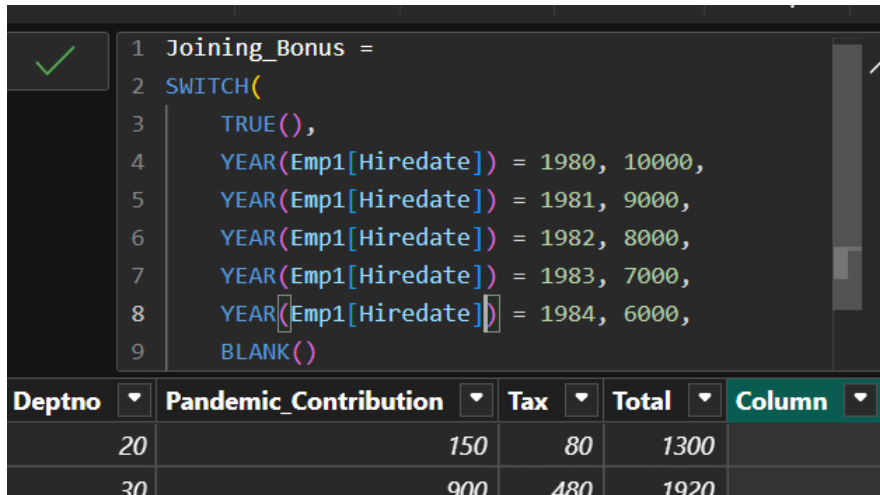


```
1 Total =
2 IF(
3     ISBLANK(Emp1[Commission]),
4     Emp1[Salary] * 1.2,
5     Emp1[Salary] + Emp1[Commission]
6 )
```

Salary	Commission	Deptno	Pandemic_Contribution	Tax	Total
800	500	20	150	80	1300
1600		30	900	480	1920
1250	200	30	400	125	1450
2975		20	1200	892.5	3570

(c) Column: Total (12 distinct values)

3. Create a column Joining_Bonus with the following conditions:
 - a. Those who joined in 1980 should get 10000 as bonus
 - b. Those who joined in 1981 should get 9000 as bonus
 - c. Those who joined in 1982 should get 8000 as bonus
 - d. Those who joined in 1983 should get 7000 as bonus
 - e. Those who joined in 1984 should get 6000 as bonus



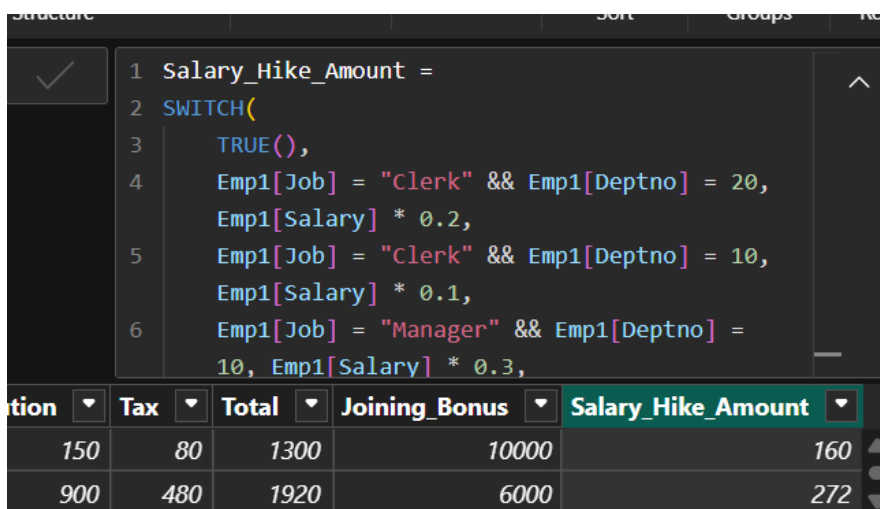
```

1 Joining_Bonus =
2 SWITCH(
3   TRUE(),
4   YEAR(Emp1[Hiredate]) = 1980, 10000,
5   YEAR(Emp1[Hiredate]) = 1981, 9000,
6   YEAR(Emp1[Hiredate]) = 1982, 8000,
7   YEAR(Emp1[Hiredate]) = 1983, 7000,
8   YEAR(Emp1[Hiredate]) = 1984, 6000,
9   BLANK()

```

Deptno	Pandemic_Contribution	Tax	Total	Column
20	150	80	1300	
30	900	480	1920	

4. Create a column Salary_Hike_Amount with the following conditions:
 - a. Employees of job type Clerk and Deptno 20 should get 20% hike
 - b. Employees of job type Clerk and Deptno 10 should get 10% hike
 - c. Employees of job type Manager and Deptno 10 should get 30% hike
 - d. Employees of job type Manager and Deptno 20 should get 15% hike
 - e. Remaining all employees should get 17% hike



```

1 Salary_Hike_Amount =
2 SWITCH(
3   TRUE(),
4   Emp1[Job] = "Clerk" && Emp1[Deptno] = 20,
5     Emp1[Salary] * 0.2,
6   Emp1[Job] = "Clerk" && Emp1[Deptno] = 10,
7     Emp1[Salary] * 0.1,
8   Emp1[Job] = "Manager" && Emp1[Deptno] =
9     10, Emp1[Salary] * 0.3,

```

Pandemic_Contribution	Tax	Total	Joining_Bonus	Salary_Hike_Amount
150	80	1300	10000	160
900	480	1920	6000	272

5. Create a column named Percentage_Contribution which will show the percentage contribution of each employee for the pandemic fund.

Structure					Sort	Groups	Relationships
✗ ✓ <pre>1 Percentage_Contribution = 2 DIVIDE(3 Emp1[Salary], 4 CALCULATE(SUM(Emp1[Salary]), ALL(Emp1)) 5) * 100 6</pre>							
ax	Total	Joining_Bonus	Salary_Hike_Amount	Column			
80	1300	10000	160				
480	1920	6000	272				
125	1450	9000	212.5				
892.5	3570	9000	446.25				

PART B - DAX Queries Measures

Total Salaries (Clerks in Dept 30 + Salesmen in Dept 20) 4700 Clerk30_Salesman20_Salary	No. of Employees in Dept 20 6 Dept20_Emp_Count	Total Salaries (Incentive > 700) 7000 High_Incentive_Salary_Total
Highest Clerk Salary in Dept 20 4400 Dept20_Clerk_MaxSalary	Total Salaries of Salesmen in Dept 20 2750 Dept20_Salesman_Salary	Total Salaries (Salary > 4400) 24K High_Salary_Total

1. Create a measure which will show the total of all salaries of employees who earn salary above 4400.

```

1 High_Salary_Total =
2 CALCULATE(
3     SUM(Emp2[SAL]),
4     FILTER(Emp2, Emp2[SAL] > 4400)
5 )
6

```

Data

Σ DEPTNO
Σ EMPNO
ENAME
☒ High_Salary_Total

	ENAME	JOB	DEPTNO	SAL	INCENTIVE
1	ROGER	ANALYST	10	Name	[High_Salary_Total]
2	ZINA	ANALYST	10		

2. Create a measure which will show the total of all the salaries of employees who earn incentives above 700.

The screenshot shows the Power BI DAX editor with the following DAX code:

```
1 High_Incentive_Salary_Total =
2 CALCULATE(
3     SUM(Emp2[SAL]),
4     FILTER(Emp2, Emp2[Incentive] > 700)
5 )
```

The Data pane on the right shows the table Emp2 with columns EMPNO, ENAME, JOB, DEPTNO, SAL, and INCENTIVE. The measure High_Incentive_Salary_Total is listed at the bottom of the Data pane.

3. Create a measure which will show the total number of employees in Deptno 20.

The screenshot shows the Power BI DAX editor with the following DAX code:

```
1 Dept20_Emp_Count =
2 CALCULATE(
3     COUNTROWS(Emp2),
4     Emp2[Deptno] = 20
5 )
```

The Data pane on the right shows the table Emp2 with columns EMPNO, ENAME, JOB, DEPTNO, SAL, and INCENTIVE. The measure Dept20_Emp_Count is listed at the bottom of the Data pane.

4. Create a measure which will add the salaries of all Salesman from Deptno 20.

The screenshot shows the Power BI DAX editor with the following DAX code:

```
1 Dept20_Salesman_Salary =
2 CALCULATE(
3     SUM(Emp2[SAL]),
4     Emp2[Deptno] = 20,
5     Emp2[Job] = "Salesman"
6 )
```

The Data pane on the right shows the table Emp2 with columns EMPNO, ENAME, JOB, DEPTNO, SAL, and INCENTIVE. The measure Dept20_Salesman_Salary is listed at the bottom of the Data pane.

5. Create a measure which will show the highest salary in job type Clerk of Deptno 20.

The screenshot shows the Power BI DAX editor with the following DAX code:

```
1 Dept20_Clerk_MaxSalary =
2 CALCULATE(
3     MAX(Emp2[SAL]),
4     Emp2[Deptno] = 20,
5     Emp2[Job] = "Clerk"
6 )
```

The Data pane on the right shows the table Emp2 with columns EMPNO, ENAME, JOB, DEPTNO, SAL, and INCENTIVE. The measure Dept20_Clerk_MaxSalary is listed at the bottom of the Data pane.

6. Create a measure which will add the salaries of all employees of Clerk in Deptno 30 as well as of Salesman in Deptno 20.

The screenshot shows the Power BI DAX editor interface. The 'Calculations' tab is active, displaying the following DAX code:

```
1 Clerk30_Salesman20_Salary =  
2 CALCULATE(  
3     SUM(Emp2[SAL]),  
4     FILTER(  
5         Emp2,  
6         (Emp2[Deptno] = 30 && Emp2[Job] =  
7             "Clerk")  
8         || (Emp2[Deptno] = 20 && Emp2[Job] =  
9             "Salesman")  
10    )  
11 )
```

On the right side, the 'Data' pane shows the 'Emp2' table expanded, listing the following measures:

- Clerk30_Salesman20_Salary (highlighted)
- Dept20_Clerk_MaxSalary
- Dept20_Emp_Count